

Please submit a pdf copy (at most 2 pages) of your own solutions to Problems 1–3, at SUCourse+ before November 13 (Thursday), 23:30.

Problem 1 (25 points) To compute the black-height of a given node in a red-black tree (RBT) in constant time, consider augmenting the black-heights of nodes as additional attributes in the nodes of the RBT. Please explain why this augmentation does not increase the asymptotic time complexity of inserting a node into an RBT in the worst case.

Problem 2 (25 points) To compute the depth of a given node in an RBT in constant time, consider augmenting the depths of nodes as additional attributes in the nodes of the RBT. Please explain by an example why this augmentation increases the asymptotic time complexity of inserting a node into an RBT in the worst case.

Problem 3 (50 points) Suppose that we perform a sequence of m insertion operations in an RBT with n nodes. Use the accounting method to determine the amortized cost per operation.

Exercise 1 (will not be graded) The dynamic set operation MIN-GAP returns the difference of the two closest numbers in a dynamic set Q of numbers. How can we augment an RBT to support the operation MIN-GAP in constant time? Is it possible to maintain this augmented information so that the running times of dynamic set operations, such as SEARCH and INSERT, on an RBT with n distinct keys are still $O(\lg(n))$? Please briefly explain.

Exercise 2 (will not be graded) Suppose that we perform a sequence of m insertion operations in an RBT with n nodes. Use the potential method to determine the amortized cost per operation.

IMPORTANT

Please pay attention to the following items about your submissions to **all** assignments in CS301.

1. Please submit **your own solutions**. Academic misconduct will not be tolerated.
2. Please submit your solutions **at SUCourse+**. Solutions emailed to us will not be graded.
3. Please submit your solutions **before the deadline**. Late submissions will not be graded.
4. Please **use an editor**, like Overleaf, to write down your solutions.
Handwritten solutions will not be graded.
5. Please submit your typeset solutions as **pdf files**.
Solutions submitted in different formats, e.g., as txt or doc files, will not be graded.
6. Please respect the specified **maximum number of pages** for submissions.
Longer submissions will not be graded.